

For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

FLUHALT FOR ORAL SUSPENSION

(Oseltamivir suspension 60 mg/ 5 ml)

COMPOSITION

Each 5 ml contains:

Oseltamivir phosphate..... 60 mg

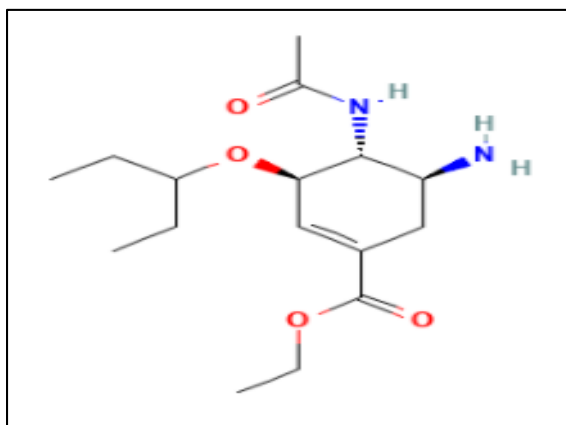
Excipients: Xanthan gum, Sodium benzoate, Saccharin sodium, Monosodium citrate, Titanium dioxide, Sorbitol, Isopropyl alcohol and Tutti-frutti Flavour

PRODUCT DESCRIPTION

Off white to light brown granular powder, forming an off white to light brown coloured suspension upon reconstitution with water. The resulting suspension has a characteristic tutti frutti flavour

DESCRIPTION

FLUHALT contains oseltamivir. Oseltamivir is an inhibitor of the influenza neuraminidase enzyme and is used as therapy and prophylaxis against influenza A and B. It is described chemically as ethyl (3R, 4R, 5S)-4-acetamido-5-amino-3-pentan-3-yloxycyclohexene-1-carboxylate. Oseltamivir has molecular formulae $C_{16}H_{28}N_2O_4$ and molecular weight is 312.40 g/mol. Oseltamivir has the following structural formula:



STRUCTURAL FORMULAE OF OSELTAMIVIR

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- Pharmacodynamics

Pharmacotherapeutic group: Antivirals for systemic use, neuraminidase inhibitors

ATC code: J05AH02.

Oseltamivir phosphate is a pro-drug of the active metabolite (oseltamivir carboxylate). The active metabolite is a selective inhibitor of influenza virus neuraminidase enzymes, which are glycoproteins found on the virion surface. Viral neuraminidase enzyme activity is important both for viral entry into uninfected cells and for the release of recently formed virus particles from infected cells, and for the further spread of infectious virus in the body.

Oseltamivir carboxylate inhibits influenza A and B neuraminidases *in vitro*. Oseltamivir phosphate inhibits influenza virus infection and replication *in vitro*. Oseltamivir given orally inhibits influenza A and B virus replication and pathogenicity *in vivo* in animal models of influenza infection at antiviral exposures similar to that achieved in man with 75 mg twice daily.

Antiviral activity of oseltamivir has been supported for influenza A and B by reported studies in healthy volunteers.

Neuraminidase enzyme IC₅₀ values for oseltamivir for clinically isolated influenza A ranged from 0.1 nM to 1.3 nM, and for influenza B was 2.6 nM. Higher IC₅₀ values for influenza B, up to a median of 8.5 nM, have been reported in published studies.

- Pharmacokinetics

General Information

Absorption

Oseltamivir is readily absorbed from the gastrointestinal tract after oral administration of oseltamivir phosphate (pro-drug) and is extensively converted by predominantly hepatic esterases to the active metabolite (oseltamivir carboxylate). At least 75 % of an oral dose reaches the systemic circulation as the active metabolite. Exposure to the pro-drug is less than 5 % relative to the active metabolite. Plasma concentrations of both pro-drug and active metabolite are proportional to dose and are unaffected by co-administration with food.

Distribution

The mean volume of distribution at steady state of the oseltamivir carboxylate is approximately 23 litres in humans, a volume equivalent to extracellular body fluid. Since neuraminidase activity is extracellular, oseltamivir carboxylate distributes to all sites of influenza virus spread.

The binding of the oseltamivir carboxylate to human plasma protein is negligible (approximately 3 %).

Biotransformation

Oseltamivir is extensively converted to oseltamivir carboxylate by esterases located predominantly in the liver. As per reported *in vitro* data, neither oseltamivir nor the active metabolite is a substrate for, or an inhibitor of, the major cytochrome P450 isoforms. No phase 2 conjugates of either compound have been reported *in vivo*.

Elimination

Absorbed oseltamivir is primarily (> 90 %) eliminated by conversion to oseltamivir carboxylate. It is not further metabolised and is eliminated in the urine. Peak plasma concentrations of oseltamivir carboxylate decline with a half-life of 6 to 10 hours in most subjects. The active metabolite is eliminated entirely by renal excretion. Renal clearance (18.8 l/h) exceeds glomerular filtration rate (7.5 l/h) indicating that tubular secretion occurs in addition to glomerular filtration. Less than 20 % of an oral radiolabelled dose is eliminated in faeces.

Other special populations

Paediatric population

Infants less than 1 year of age: The rate of clearance of the active metabolite, corrected for body-weight, have been reported to decrease with ages in influenza infected children below one year. Metabolite exposures are also more variable in the youngest infants. The reported data indicates that the exposure following a 3 mg/kg dose in infants 0-12 months of age provides pro-drug and metabolite exposures anticipated to be efficacious with a safety profile comparable to that reported in older children and adults using the approved dose (see INDICATIONS and DOSE AND METHOD OF ADMINISTRATION). The reported adverse events have been consistent with the established safety profile in older children.

No data has been reported for infants below 1 year of age for post exposure prevention of influenza. Prevention during an influenza epidemic in the community has not been reported in children below 12 years of age.

Post-exposure prevention of influenza in infants less than 1 year of age during a pandemic: Simulation of once daily dosing of 3 mg/kg in infants < 1 year reported an exposure in the same range or higher than for once daily dosing of 75 mg in adults. Exposure does not exceed that for treatment of infants < 1 year (3 mg/kg twice daily) and is anticipated to result in a comparable safety profile (see SIDE EFFECTS/UNDESIRABLE EFFECTS). No studies of prophylaxis in infants aged < 1 have been reported.

Infants and children 1 year of age or older: Younger children have been reported to clear both the pro-drug and its active metabolite faster than adults, resulting in a lower exposure for a given mg/kg dose. Doses of 2 mg/kg give oseltamivir carboxylate exposures comparable to those achieved in adults receiving a single 75 mg dose (approximately 1 mg/kg). The pharmacokinetics of oseltamivir in children and adolescents 12 years of age or older have been reported to be similar to those in adults.

Elderly

Exposure to the active metabolite at steady state has been reported to be 25 to 35 % higher in older people (age 65 to 78 years) compared to adults less than 65 years of age given comparable doses of oseltamivir. Half-lives reported in older people were similar to those reported in young adults. On the basis of drug exposure and tolerability, dosage adjustments are not required for older people unless there is evidence of moderate or severe renal impairment (creatinine clearance below 60 ml /min) (see DOSE AND METHOD OF ADMINISTRATION).

Renal impairment

Administration of 100 mg oseltamivir phosphate twice daily for 5 days to patients with various degrees of renal impairment reported that exposure to oseltamivir carboxylate is inversely proportional to declining renal function. For dosing, see DOSE AND METHOD OF ADMINISTRATION.

Hepatic impairment

Exposure to oseltamivir is not expected to be increased significantly nor is exposure to the active metabolite expected to be significantly decreased in patients with hepatic impairment (see DOSE AND METHOD OF ADMINISTRATION).

Pregnant Women

The Oseltamivir dosage regimen described in DOSE AND METHOD OF ADMINISTRATION reportedly results in lower exposure (30% on average across all trimesters) to the active metabolite in pregnant women compared to non-pregnant women. The lower predicted exposure however, remains above inhibitory concentrations (IC₉₅ values) and at a therapeutic level for a range of influenza virus strains. In addition, reportedly there is evidence that shows benefit of the current dosing regimen in this patient population. Therefore, dose adjustments are not recommended for pregnant women in the treatment or prophylaxis of influenza (see USE IN SPECIAL POPULATIONS).

Immunocompromised Patients

Treatment of adult and paediatric (< 18 years old) immunocompromised patients with oseltamivir (as described in DOSE AND METHOD OF ADMINISTRATION) has been reported to result in an increased predicted exposure (from approximately 5% up to 50%) to the active metabolite when compared to non-immunocompromised patients with comparable creatinine clearance. Due to the wide safety margin of the

active metabolite, no dose adjustments are required in patients due to their immunocompromised status. However, for immunocompromised patients with renal impairment, doses should be adjusted as outlined in DOSE AND METHOD OF ADMINISTRATION.

Reportedly there is no meaningful additional benefit in exposures higher than those achieved after the administration of the standard dose in immunocompromised patients.

INDICATIONS

Treatment of influenza

Oseltamivir is indicated for the treatment of influenza in children 6 to 12 months of age during a pandemic influenza outbreak.

Oseltamivir is indicated for the treatment of influenza in adults and children ≥ 1 year of age (see *DOSE AND METHOD OF ADMINISTRATION, WARNINGS AND PRECAUTIONS*).

Treatment should commence as soon as possible but no later than forty-eight hours after the onset of the initial symptoms of infection.

Prophylaxis of influenza

Oseltamivir is indicated for post exposure prophylaxis in adults and children ≥ 1 year following contact with a clinically diagnosed influenza case when influenza virus is circulating in the community.

DOSE AND METHOD OF ADMINISTRATION

FLUHALT is available at the strength of 60 mg/ 5 ml only and may not be suitable for all the dosing recommendations mentioned below. In such cases, other approved strengths should be used.

DOSE

75 mg doses can be administered as one 75 mg capsule.

Adults, and adolescents 13 years and over

Treatment: The recommended oral dose is 75 mg oseltamivir twice daily for 5 days for adolescents (13 to 17 years of age) and adults.

Body Weight	Recommended dose for 5 days	Recommended dose for 10 days* Immunocompromised Patients
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> 40 kg	75 mg twice daily	75 mg twice daily
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* The recommended treatment duration in immunocompromised adults and adolescents is 10 days.

Treatment should be initiated as soon as possible within the first two days of onset of symptoms of influenza.

Post-exposure prevention: The recommended dose for prevention of influenza following close contact with an infected individual is 75 mg oseltamivir once daily for 10 days for adolescents (13 to 17 years of age) and adults.

Body Weight	Recommended dose for 10 days	Recommended dose for 10 days Immunocompromised Patients
> 40 kg	75 mg once daily	75 mg once daily

Therapy should begin as soon as possible within two days of exposure to an infected individual.

Prevention during an influenza epidemic in the community: The recommended dose for prevention of influenza during a community outbreak is 75 mg oseltamivir once daily for up to 6 weeks (or up to 12 weeks in immunocompromised patients, see WARNINGS AND PRECAUTIONS, SIDE EFFECTS/UNDESIRABLE EFFECTS and Pharmacodynamics).

Paediatric population

Children 1 to 12 years of age

Treatment: The following weight-adjusted dosing regimens are recommended for treatment of infants and children 1 year of age or older:

Body Weight	Recommended dose for 5 days	Recommended dose for 10 days* Immunocompromised Patients
10 kg to 15 kg	30 mg twice daily	30 mg twice daily
> 15 kg to 23 kg	45 mg twice daily	45 mg twice daily
> 23 kg to 40 kg	60 mg twice daily	60 mg twice daily
> 40 kg	75 mg twice daily	75 mg twice daily

*The recommended treatment duration in immunocompromised children (≥ 1 year old) is 10 days.

Treatment should be initiated as soon as possible within the first two days of onset of symptoms of influenza.

Post-exposure prevention: The recommended post-exposure prevention dose of oseltamivir is:

Body Weight	Recommended dose for 10 days	Recommended dose for 10 days For Immunocompromised Patients
10 kg to 15 kg	30 mg once daily	30 mg once daily
> 15 kg to 23 kg	45 mg once daily	45 mg once daily
> 23 kg to 40 kg	60 mg once daily	60 mg once daily
> 40 kg	75 mg once daily	75 mg once daily

Prevention during an influenza epidemic in the community: Prevention during an influenza epidemic has not been reported in children below 12 years of age.

Infants 6 – 12 months of age

Treatment: The recommended treatment dose for infants 6 - 12 months of age is 3 mg/kg twice daily. The following dosing regimen is recommended for treatment of infants 6 - 12 months of age:

Body weight*	Recommended dose for 5 days	Recommended dose for 10 days** Immunocompromised Patients
3 kg	9 mg twice daily	9 mg twice daily
4 kg	12 mg twice daily	12 mg twice daily
5 kg	15 mg twice daily	15 mg twice daily
6 kg	18 mg twice daily	18 mg twice daily
7 kg	21 mg twice daily	21 mg twice daily
8 kg	24 mg twice daily	24 mg twice daily
9 kg	27 mg twice daily	27 mg twice daily
10 kg	30 mg twice daily	30 mg twice daily

* This table is not intended to contain all possible weights for this population. For all patients under the age of 1 year, 3 mg/kg should be used to determine dose regardless of the weight of the patient.

Treatment should be initiated as soon as possible within the first two days of onset of symptoms of influenza.

** The recommended duration in immunocompromised infants (6-12 months old) is 10 days.

This dosing recommendation is not intended for premature infants, i.e. those with a post-conceptual age less than 36 weeks. Insufficient data have been reported for these patients, in whom different dosing may be required due to the immaturity of physiological functions.

Special populations

Hepatic impairment

No dose adjustment is required either for treatment or for prevention in patients with hepatic dysfunction. No studies have been reported in paediatric patients with hepatic disorder.

Renal impairment

Treatment of influenza: Dose adjustment is recommended for adults and adolescents (13 to 17 years of age) with moderate or severe renal impairment. Recommended doses are detailed in the table below.

Creatinine clearance	Recommended dose for treatment
> 60 (ml/min)	75 mg twice daily
> 30 to 60 (ml/min)	30 mg (suspension or capsules) twice daily
> 10 to 30 (ml/min)	30 mg (suspension or capsules) once daily
≤10 (ml/min)	Not recommended (no data available)
Haemodialysis patients	30 mg after each haemodialysis session

Peritoneal dialysis patients*	30 mg (suspension or capsules) single dose
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* Data derived from reported studies in continuous ambulatory peritoneal dialysis (CAPD) patients; the clearance of oseltamivir carboxylate is expected to be higher when automated peritoneal dialysis (APD) mode is used. Treatment mode can be switched from APD to CAPD if considered necessary by a nephrologist.

Prevention of influenza: Dose adjustment is recommended for adults and adolescents (13 to 17 years of age) with moderate or severe renal impairment as detailed in the table below.

Creatinine clearance	Recommended dose for prevention
> 60 (ml/min)	75 mg once daily
> 30 to 60 (ml/min)	30 mg (suspension or capsules) once daily
> 10 to 30 (ml/min)	30 mg (suspension or capsules) every second day
≤10 (ml/min)	Not recommended (no data available)
Haemodialysis patients	30 mg after every second haemodialysis session
Peritoneal dialysis patients*	30 mg (suspension or capsules) once weekly

* Data derived from reported studies in continuous ambulatory peritoneal dialysis (CAPD) patients; the clearance of oseltamivir carboxylate is expected to be higher when automated peritoneal dialysis (APD) mode is used. Treatment mode can be switched from APD to CAPD if considered necessary by a nephrologist.

There is insufficient data reported in infants and children (12 years of age and younger) with renal impairment to be able to make any dosing recommendation.

Elderly

No dose adjustment is required, unless there is evidence of moderate or severe renal impairment.

Immunocompromised patients

Treatment: For treatment of influenza, the recommended duration for immunocompromised patients is 10 days (see WARNINGS AND PRECAUTIONS, SIDE EFFECTS/UNDESIRABLE EFFECTS and Pharmacodynamics). No dose adjustment is necessary. Treatment should be initiated as soon as possible within the first two days of onset of symptoms of influenza.

Seasonal prophylaxis: Longer duration of seasonal prophylaxis up to 12 weeks has been reported in immunocompromised patients (see WARNINGS AND PRECAUTIONS, SIDE EFFECTS/UNDESIRABLE EFFECTS and Pharmacodynamics).

Dose modifications

Oral use.

Patients who are unable to swallow capsules may receive appropriate doses of oseltamivir suspension.

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients of the product.

WARNINGS AND PRECAUTIONS

Oseltamivir is effective only against illness caused by influenza viruses. No data has been reported for efficacy of oseltamivir in any illness caused by agents other than influenza viruses (see Pharmacodynamics).

Oseltamivir is not a substitute for influenza vaccination. Use of oseltamivir must not affect the evaluation of individuals for annual influenza vaccination. The protection against influenza lasts only as long as oseltamivir is administered. Oseltamivir should be used for the treatment and prevention of influenza only when reliable epidemiological data indicate that influenza virus is circulating in the community.

Susceptibility of circulating influenza virus strains to oseltamivir has been reported to be highly variable (see Pharmacodynamics). Therefore, prescribers should take into account the most recent information available on oseltamivir susceptibility patterns of the currently circulating viruses when deciding whether to use oseltamivir.

Severe concomitant condition

No information is available regarding the safety and efficacy of oseltamivir in patients with any medical condition sufficiently severe or unstable to be considered at imminent risk of requiring hospitalization.

Immunocompromised patients

The efficacy of oseltamivir in either treatment or prophylaxis of influenza in immunocompromised patients has not been firmly established (see Pharmacodynamics).

Cardiac / respiratory disease

Efficacy of oseltamivir in the treatment of subjects with chronic cardiac disease and/or respiratory disease has not been reported. No difference in the incidence of complications has been reported between the treatment and placebo groups in this population (see Pharmacodynamics).

Paediatric population

No data has been reported to allow a dose recommendation for premature children (< 36 weeks post-conceptual age)

Severe renal impairment

Dose adjustment is recommended for both treatment and prevention in adolescents (13 to 17 years of age) and adults with severe renal impairment. There is insufficient reported data in infants and children (1 year of

age or older) with renal impairment to be able to make any dosing recommendation (see DOSE AND METHOD OF ADMINISTRATION and Pharmacokinetics).

Neuropsychiatric events

Neuropsychiatric events have been reported during administration of oseltamivir in patients with influenza, especially in children and adolescents. These events are also experienced by patients with influenza without oseltamivir administration. Patients should be closely monitored for behavioral changes, and the benefits and risks of continuing treatment should be carefully reported for each patient (see SIDE EFFECTS/UNDESIRABLE EFFECTS).

FLUHALT contains sorbitol, which may cause gastrointestinal discomfort and mild laxative effect.

Effects on ability to drive and use machines

Oseltamivir has no influence on the ability to drive and use machines.

DRUG INTERACTIONS

Pharmacokinetic properties of oseltamivir, such as low protein binding and metabolism independent of the CYP450 and glucuronidase systems (see Pharmacokinetics), suggest that clinically significant drug interactions via these mechanisms are unlikely.

Probenecid

No dose adjustment is required when co-administering with probenecid in patients with normal renal function. Co-administration of probenecid, a potent inhibitor of the anionic pathway of renal tubular secretion, results in an approximate 2-fold increase in exposure to the active metabolite of oseltamivir.

Amoxicillin

Oseltamivir has no kinetic interaction with amoxicillin, which is eliminated via the same pathway, suggesting that oseltamivir interaction with this pathway is weak.

Renal elimination

Clinically important drug interactions involving competition for renal tubular secretion are unlikely, due to the known safety margin for most of these substances, the elimination characteristics of the active metabolite (glomerular filtration and anionic tubular secretion) and the excretion capacity of these pathways. However, care should be taken when prescribing oseltamivir in subjects when taking co-excreted agents with a narrow therapeutic margin (e.g. chlorpropamide, methotrexate, phenylbutazone).

Additional information

No pharmacokinetic interactions between oseltamivir or its major metabolite have been reported when co-administering oseltamivir with paracetamol, acetylsalicylic acid, cimetidine, antacids (magnesium and aluminium hydroxides and calcium carbonates), rimantadine or warfarin (in subjects stable on warfarin and without influenza).

SIDE EFFECTS/UNDESIRABLE EFFECTS

Summary of the safety profile

In adults/adolescents, the most commonly reported adverse reactions (ARs) were nausea and vomiting. The majority of these ARs were reported on a single occasion on either the first or second treatment day and resolved spontaneously within 1-2 days. In children, the most commonly reported adverse reaction has been reported to be vomiting. In the majority of patients, these ARs did not lead to discontinuation of oseltamivir.

The following serious adverse reactions have been rarely reported since oseltamivir has been marketed: Anaphylactic and anaphylactoid reactions, hepatic disorders (fulminant hepatitis, hepatic function disorder and jaundice), angioneurotic oedema, Stevens-Johnson syndrome and toxic epidermal necrolysis, gastrointestinal bleeding and neuropsychiatric disorders.

(Regarding neuropsychiatric disorders, see WARNINGS AND PRECAUTIONS)

Tabulated list of adverse reactions

Treatment and prevention of influenza in adults and adolescents:

The most frequently ARs reported at the recommended dose (75 mg bid for 5 days for treatment and 75 mg od for up to 6 weeks for prophylaxis) are shown in table below.

The safety profile reported in patients who received the recommended dose of oseltamivir for prophylaxis (75 mg once daily for up to 6 weeks) has been reported to be qualitatively similar to that reported in patients receiving oseltamivir for treatment, despite a longer duration of dosing.

Table: Adverse reactions reported with oseltamivir for treatment and prevention of influenza in adults and adolescents

System (SOC)	Organ	Class	Adverse reactions according to frequency			
			Very common	Common	Uncommon	Rare

Infections and infestations		Bronchitis, Herpes simplex, Nasopharyngitis, Upper respiratory tract infections, Sinusitis		
Blood and lymphatic system disorders				Thrombocytopenia
Immune system disorders			Hypersensitivity reaction	Anaphylactic reactions, Anaphylactoid reactions
Psychiatric disorders				Agitation, Abnormal behaviour, Anxiety, Confusion, Delusions, Delirium, Hallucination, Nightmares, Self-injury
Nervous system disorders	Headache	Insomnia	Altered level of consciousness, Convulsion	
Eye disorders				Visual disturbance
Cardiac disorders			Cardiac arrhythmia	
Respiratory, thoracic and mediastinal disorders		Cough, Sore throat, Rhinorrhea		
Gastrointestinal disorders	Nausea	Vomiting Abdominal pain (incl. upper abdominal pain), Dyspepsia		Gastrointestinal bleedings, Haemorrhagic colitis
Hepatobiliary disorders			Elevated liver enzymes	Fulminant hepatitis, Hepatic failure, Hepatitis
Skin and subcutaneous tissue disorders			Eczema, Dermatitis, Rash, Urticaria	Angioneurotic oedema, Erythema multiforme, Stevens-Johnson syndrome, Toxic epidermal necrolysis
General disorders and administration site conditions		Pain Dizziness (incl. vertigo), Fatigue, Pyrexia, Pain in limb		

Table below shows the most frequently reported ARs in paediatric patients.

Table: Adverse reactions reported with oseltamivir for treatment and prevention of influenza in children (age/weight-based dosing [30 mg to 75 mg o.d.]

System (SOC)	Organ	Class	Adverse reactions according to frequency			
			Very common	Common	Uncommon	Rare

Infections and infestations		Otitis media,		
Nervous system disorders		Headache		
Eye disorders		Conjunctivitis (including red eyes, eye discharge and eye pain)		
Ear and labyrinth disorders		Earache	Tympanic membrane disorder	
Respiratory, thoracic and mediastinal disorders	Cough, Nasal congestion	Rhinorrhoea		
Gastrointestinal disorders	Vomiting	Abdominal pain (incl. upper abdominal pain), Dyspepsia, Nausea		
Skin and subcutaneous tissue disorders			Dermatitis (including allergic and atopic dermatitis)	

Description of selected adverse reactions

Psychiatric disorders and nervous system disorders

Influenza can be associated with a variety of neurologic and behavioral symptoms which can include events such as hallucinations, delirium, and abnormal behavior, in some reported cases resulting in fatal outcomes. These events may occur in the setting of encephalitis or encephalopathy but can occur without obvious severe disease.

In patients with influenza who were receiving oseltamivir, there have been post marketing reports of convulsions and delirium (including symptoms such as altered level of consciousness, confusion, abnormal behaviour, delusions, hallucinations, agitation, anxiety, nightmares), in a very few reported cases resulting in self-injury or fatal outcomes. These events have been reported primarily among paediatric and adolescent patients and often had an abrupt onset and rapid resolution. The contribution of oseltamivir to those events is unknown. Such neuropsychiatric events have also been reported in patients with influenza who were not taking oseltamivir.

Hepato-biliary disorders

Hepato-biliary system disorders, including hepatitis and elevated liver enzymes in patients with influenza-like illness. These cases include fatal fulminant hepatitis/hepatic failure.

Other special populations

Paediatric population (infants less than one year of age)

Safety profile of oseltamivir therapy in influenza infected children less than one year of age, has been reported to be similar among age cohorts with vomiting, diarrhoea and diaper rash being the most frequently reported adverse events (see Pharmacokinetics). Insufficient data have been reported for infants who have a post-conceptual age of less than 36 weeks.

The safety profile of oseltamivir administered for treatment of influenza in infants less than one year of age has been reported to be similar to the established safety profile of children aged one year and older.

Older people and patients with chronic cardiac and/or respiratory disease

In general, the safety profile in the patients “at risk” (patients at higher risk of developing complications associated with influenza, e.g. older people and patients with chronic cardiac or respiratory disease) has been reported to be qualitatively similar to that in otherwise healthy adults/adolescents.

Immunocompromised patients

The most frequent adverse reaction reported in immunocompromised children receiving standard dose or high dose regimens (double dose or triple dose) of oseltamivir for treatment of influenza was vomiting (28%).

Children with pre-existing bronchial asthma

In general, the adverse reaction profile in children with pre-existing bronchial asthma has been reported to be qualitatively similar to that of otherwise healthy children.

USE IN SPECIAL POPULATIONS

Pregnancy

Influenza is associated with adverse pregnancy and foetal outcomes, with a risk of major congenital malformations, including congenital heart defects. A large amount of reported data on oseltamivir exposure of pregnant women from post-marketing reports and reported studies (indicate no malformative nor feto/neonatal toxicity by oseltamivir.

The rate of major congenital heart defects following oseltamivir exposure during the first trimester has been reported to be 1.76% compared to 1.01% in unexposed pregnancies from the general population (Odds Ratio 1.75, 95% Confidence Interval 0.51 to 5.98). The clinical significance of this finding is not clear. Additionally, this reported data was too small to reliably assess individual types of major malformations; moreover women exposed to oseltamivir and women unexposed could not be made fully comparable, in particular whether or not they had influenza.

Reproductive toxicity has not been reported in animals.

The use of oseltamivir may be considered during pregnancy if necessary and after considering the available safety and benefit information, and the pathogenicity of the circulating influenza virus strain.

Breastfeeding

In lactating rats, oseltamivir and the active metabolite are excreted in milk. Very limited information has been reported on children breast-fed by mothers taking oseltamivir and on excretion of oseltamivir in breast milk. Limited data reported that oseltamivir and the active metabolite have been reported to be detected in breast milk, however the levels were low, which would result in a subtherapeutic dose to the infant. Considering this information, the pathogenicity of the circulating influenza virus strain and the underlying condition of the breastfeeding woman, administration of oseltamivir may be considered, where there are clear potential benefits to breastfeeding mothers.

Fertility

Reportedly there is no evidence that oseltamivir has an effect on male or female fertility.

OVERDOSE

In the majority of cases reporting overdose, no adverse events have been reported.

Adverse events reported following overdose were similar in nature and distribution to those reported with therapeutic doses of oseltamivir, described in SIDE EFFECTS/UNDESIRABLE EFFECTS.

No specific antidote is known.

Paediatric population

Overdose has been reported more frequently for children than adults and adolescents. Caution should be exercised when preparing oseltamivir oral suspension and when administering oseltamivir products to children.

STORAGE

Before Mixing

Store below 30°C, protected from moisture.

Direction for Mixing:

Loosen powder. Add boiled and cooled water slowly upto the arrow mark and shake vigorously. Adjust the volume upto the arrow mark by adding more water, if necessary.

SHAKE WELL BEFORE USE

Store the prepared suspension under refrigerator (2°C – 8°C)

Use the prepared suspension within 14 days

PACKING

25 ml, 60 ml, 75 ml and 100 ml HDPE bottles

Date of Revision: March 2022.

MANUFACTURED BY AND PRODUCT REGISTRATION HOLDER:

Ranbaxy (Malaysia) Sdn. Bhd.

Lot 23, Bakar Arang Industrial Estate

08000 Sungai Petani

Kedah, Malaysia